Question ID: a87c3925

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Craft and Structure	Cross-Text Connections	Hard

Question
Text 1

Soy sauce, made from fermented soybeans, is noted for its umami flavor. Umami—one of the five basic tastes along with sweet, bitter, salty, and sour—was formally classified when its taste receptors were discovered in the 2000s. In 2007, to define the pure umami flavor scientists Rie Ishii and Michael O'Mahony used broths made from shiitake mushrooms and kombu seaweed, and two panels of Japanese and US judges closely agreed on a description of the taste.

Text 2

A 2022 experiment by Manon Jünger et al. led to a greater understanding of soy sauce's flavor profile. The team initially presented a mixture of compounds with low molecular weights to taste testers who found it was not as salty or bitter as real soy sauce. Further analysis of soy sauce identified proteins, including dipeptides, that enhanced umami flavor and also contributed to saltiness. The team then made a mix of 50 chemical compounds that re-created soy sauce's flavor.

Based on the texts, if Ishii and O'Mahony (Text 1) and Jünger et al. (Text 2) were aware of the findings of both experiments, they would most likely agree with which statement?

Answer
A. On average, the diets of people in the United States tend to have fewer foods that contain certain dipeptides than the diets of people in Japan have.

B. Chemical compounds that activate both the umami and salty taste receptors tend to have a higher molecular weight than those that only activate umami taste receptors.

C. Fermentation introduces proteins responsible for the increase of umami flavor in soy sauce, and those proteins also increase the perception of saltiness.

D. The broths in the 2007 experiment most likely did not have a substantial amount of the dipeptides that played a key part in the 2022 experiment.

Correct Answer: D

Rationale
Choice D is the best answer. Ishii and O'Mahony were trying to isolate the pure umami flavor, while Jünger was trying to recreate soy sauce, which has a mix of flavors that includes umami. Accordingly, the broths from Text 1 are not described as having any soy sauce in them—just "shiitake mushrooms and kombu seaweed." So they probably don't have as much of the dipeptides described in Text 2, which were found to be a key part of soy sauce's umami-ness and its saltiness.

Choice A is incorrect. Neither text supports this. Neither text gets into the diets of people in the United States, nor the diets of people in Japan. Choice B is incorrect. Neither text supports this. Text 2 does talk about the molecular weights of chemical compounds, but there isn't enough information provided about molecular weights in Text 1 to make an inference about what the scientists in Text 1 would say. Choice C is incorrect. Neither text supports this. Text 1 briefly mentions that soy sauce is "made from fermented soybeans," but it never claims that fermentation is responsible for its flavor in any way. And Text 2 never mentions fermentation at all.